Swechha Pandey

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Senior Software Development Engineer(SDE) Franklin,TN, USA www.linkedin.com/in/swechha-pandey

Summary

Seasoned Software Development Engineering Lead with over a decade of experience ensuring quality for software, particularly for cutting-edge machine learning and AI projects.

Adept at designing, developing, and implementing robust automated testing frameworks using Python, Selenium WebDriver, and Robot Framework. Profcient in Python libraries like Pandas, NumPy, scikit-learn, and PySpark,
demonstrating expertise in data manipulation, analysis, and machine learning, which I leveraged to develop data-driven test automation solutions that improved test coverage by 25%.

Experienced in leveraging multiple cloud platforms like AWS, Azure, and GCP, as well as containerization technologies like Kubernetes to build scalable testing solutions. At Smarsh Inc., contributed to an ML/NLP surveillance platform by designing and developing automated test scripts that resulted in a 15% reduction in testing time and improved defect detection rates. This experience involved collaborating with cross-functional teams and ensuring rigorous testing coverage for microservices, contributing to a 20% increase in deployment frequency.

Successfully led and mentored teams of test engineers, fostered collaborative environments, and implemented best practices for automation, performance, and integration testing, especially in microservices environments. These improvements resulted in a 10% reduction in production defects. Strong understanding of Agile and Scrum methodologies.
Proven ability to analyze test results, provide insights, and drive data-driven decisions for continuous improvement.

Skills

Python

Cloud -AWS, Azure, GCP

Machine Learning

API Testing

API Security Testing (using Oauth etc.)

Java

Docker

Kubernetes

Linux

UI testing

Robot framework

Jira

Git

Agile Methodologies

Test Management

Test Planning & Estimation

CI/CD

BDD

TDD

Employment History

Senior Software Development Engineer(SDE), Smarsh Inc
Dec 2020 — Present, Franklin, TN

- Developed and maintained a robust and scalable Python-based test automation framework usingpytest, various python libaries like Pandas, and NumPy to validate the functionality and performance of the ML/NLP surveillance platform.
- Implemented a comprehensive suite of integration tests using pytest, leveraging fxture mechanisms to create reusable components, mock external dependencies, and set up complex test environments, thereby ensuring seamless communication between microservices and improving overall platform stability and maintainability.
- Collaborated with model developers and the validation team to enhance the testing framework with machine learning models, improving anomaly detection accuracy and eficiency.
- Automated the analysis of large datasets generated by the surveillance platform using Python to ensure data quality, resulting in a 20% reduction in testing time.
- Familiarity with Machine learning LLM models and advance framework like Deep learning• Led and mentored a team of test engineers, championing best practices for test automation, performance testing, and integration testing within a microservices environment to foster a culture of innovation, collaboration, and continuous improvement.
- Analyze test results, distilling meaningful insights and actionable recommendations for stakeholders, and drive data-driven decision-making to fuel continuous improvement initiatives.
- Used AI/ML frameworks for Building, Training and Testing Machine Learning Models.

Technical Test Lead, eviCore inc.
Aug 2019 — Dec 2020 Franklin, TN

- Recognized for the excellent work by client and stakeholders for Quarterly standout work in improving eficiency and quality of framework resulting into cost optimization and time to release reduction.

SQL

LinkedIn

MySQL

Selenium

TestNG

Maven

- Lead the delivery and support of automation solutions, ensuring continuousimprovement and eficiency gains.
- Build CI/CD pipelines specifcally tailored for Automation testing to enable rapid and reliable software releases.
- Author and update manual test cases, optimizing organization and productivity for the team.
- Collaborate within Agile and Scrum methodologies to achieve project milestones and meet tight deadlines.
- Coordinate with of-site teams to execute tests and facilitate seamless product releases.
- Actively participate in scrum ceremonies such as sprint planning, backlog grooming, sprint review, and retrospective meetings.
- Evaluate test results, generate detailed reports, and provide actionable insights to development teams for continuous improvement.

Test Lead, Citiustech
Aug 2015 — Dec 2019 Mumbai, India

- Analyze and clarify design requirements, translating them into comprehensive test strategies, plans, and scenarios.
- Develop and execute test cases for Smoke, Functional, and Regression testing to validate microservices functionality.
- Implement and manage periodic regression runs to maintain product quality and stability.
- Design and develop an Automation framework for UI testing, utilizing Selenium WebDriver with C# and Nunit framework.

- Analyzed and documented software requirements to create a comprehensive test plan
- Execute tests rigorously, report and track bugs to closure promptly.
- Develop Automation Test scripts using Selenium WebDriver, Java, TestNG, Maven, and follow the Page Object Model for maintainable automation.
- Conduct weekly regression runs and promptly address maintenance issues.
- Maintain and optimize automation test scripts for comprehensive test coverage.
- Provide accurate estimates for testing phases from planning to regression.
- Collaborate closely with teams to ensure high-quality product delivery.
- Enforce strict adherence to company standards in automated testing.
- Communicate efectively with developers and quality teams to resolve defects eficiently.
- Led a cross-functional team of testers and developers to quickly identify and resolve critical bugsbefore release
- Created and maintained a comprehensive suite of regression and functional tests that ensured high software quality.

Software QA Engineer, Cybage Software Pvt Ltd. Feb 2013 — Aug 2015 Pune, India

- Create and execute test plans to ensure software meets requirements.
- Conduct manual and automated testing to identify defects and improve quality.
- Collaborate with developers and product managers to clarify requirements and resolve issues.
- Document and report bugs, ensuring accurate tracking and resolution.
- Develop and maintain automated test scripts using Selenium WebDriver and TestNG to streamline testing eforts.
- Perform functional and regression testing for applications built on the .NET framework.
- Verify that software complies with industry standards and organizational guidelines.
- Analyze and optimize existing test cases to improve coverage and eficiency.
- Continuously evaluate and enhance testing processes to ensure high-quality software delivery.

Certifcates

Certifcation in Generative AI (April 2024 – Present)

DeepLearning.AI Certifcation in,
Natural Language Processing with Classifcation and Vector Space (Dec 2024 – Present)

University of Michigan Certifcation in,
Python Data Structures (Dec 2024 – Present)

IBM Certifcation in
Python For Data Science, AI & Development (Jan 2025 - Present)

Education

Pune University, Bachelor of Engineering Aug
2008 — Jun 2012 Pune

CDAC, Diploma in Advance Computing
Aug 2012 — Feb 2013 Navi Mumbai